THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DAR ES SALAAM REGION

FORM FOUR - MOCK EXAMINATION - 2026

PHYSICS 2A (ACTUAL PRACTICAL)

CODE:031/2A
TIME:2:30Hours **Wednesday 20th May, 2026 A.M**

INSTRUCTIONS

1. This paper consists of two (2) questions.
2. Answer all questions.
3. Each question carries twenty **five (25)** marks.
4. Non-programmable calculators may be used.
5. Mobile phones, smart watches and any other unauthorized gadgets are **not** allowed in the examination room.

1. The aim of the experiment was to determine the density of a meter rule
Proceed as follows:

(a) Locate the centre of gravity C of the meter rule EF by balancing it freely on the triangular prism/knife edge.

(b) Suspend 100 g mass on the ruler with distance d = 10 cm from C, adjust the position of the prism to get a balance as shown in the following figure.

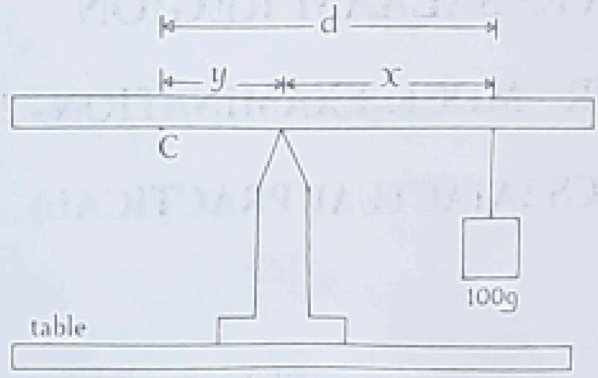

figure 1

(c) Record the distance y from the centre of gravity to the prism and distance x from prism to the known mass of 100 g

(d) Repeat the procedures in 1(b) and (c) by increasing the distance of 100 g to d = 15 cm, 20 cm, 25 and 30 cm

Questions

(i) Tabulate the results in a suitable table showing the values of d, x and y.

(ii) Plot a graph of y (cm) against x (cm).

(iii) Determine the slope of the graph.

(iv) Measure and record the length, width and the thickness of the meter rule provided.

(v) Determine the density of the meter rule and mass of a meter rule.

(vi) Describe how the slope obtained from the graph is related to the mass of the meter rule. **(25 marks)**

2. The aim of this experiment is to determine the refractive index η of a given glass block

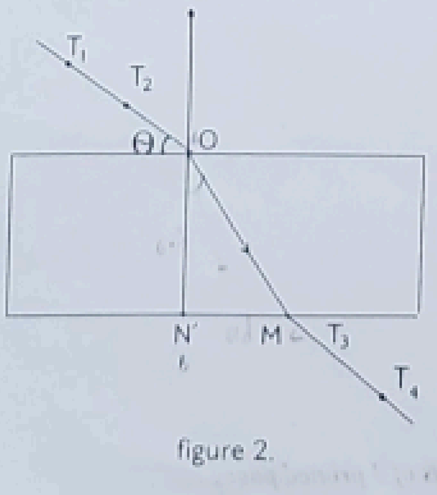

figure 2.

(a) Place the glass block on the white paper on a drawing board. Using a pencil, trace the outline of the block.

(b) Remove the glass block and draw the normal line NO near the left end of the block.

(c) Using a protractor and pencil measure the angle 20° with the surface RR of the block, erect two pins T_1 and T_2 as shown above.

(d) Replace the glass block to its holder sheet of paper.

(e) Stick the pins T_3 and T_4 at a position such that they are in straight line with pins T_1 and T_2 as seen through the block. Remove the block and draw a complete path of the ray.

(f) Measure the length MN' and ON'.

(g) Repeat the procedures for the values of $\theta = 30°, 40°, 50°$ and $60°$ respectively. In each case make a drawing on a fresh page of drawing paper.

Questions

(i) Record the value of θ, MN', ON', $\frac{MN'}{ON'}$ and cosθ

(ii) Plot the graph of $\frac{MN'}{ON'}$ against cos θ

(iii) Find the slope G of the graph

(iv) Calculate the refractive index η given that $G = \frac{1}{\eta}$

(v) State two sources of errors